SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

MCCB 400AF (MCCB-ABS-403c) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 50 mm above and below the breaker for arc gases; 10 mm between adjacent breakers
- Operating temperature: -5 to +40 °C (derating above 40 °C)
- Degree of protection: IP20 (front IP40)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: 18 N·m (M10)
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check the rating plate against the single-line diagram
2. Verify tightening torque of all terminals
3. Measure insulation resistance (≥ 5 MΩ at 500 V DC)
4. Operate the breaker manually ON/OFF three times
5. Check trip unit settings against the coordination study

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Breaker trips at motor start	Motor inrush exceeds magnetic trip setting	TG-MCCB-ABS-403c §3.1
Terminal overheating / discoloration	Loose connection or undersized cable	TG-MCCB-ABS-403c §3.2
Breaker cannot be reset after trip	Handle not moved to reset position before ON	TG-MCCB-ABS-403c §3.3

On-site commissioning support can be requested through the Global CS Center. For low-voltage circuit breakers the request is handled by the technical support team.